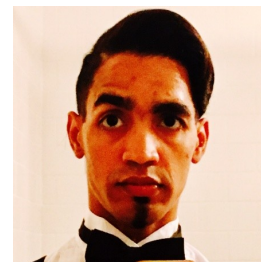


# Piter Z. Garcia Bautista

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## Research Profile

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Graduate researcher focused on deep learning and computational biology, with hands-on experience in neural network-based approaches to biological sequence and structure data. Background includes RNA secondary structure prediction with thermodynamic deep learning enhancements (BIO614), RNA-seq differential-expression analysis (BIO550), fairness-aware ML applied to bioinformatics diagnostics (ISTE-780), and reproducible evaluation pipeline design for biological data. Current Graduate Assistant work (Quantum and AI Research, RIT) deepens expertise in neural architectures, uncertainty quantification, and structured experimental methodology. Motivated to work on deep learning methods that unlock biological insights from complex, high-dimensional, and noisy molecular data.

## Research Interests

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Deep learning for biological sequence and structure analysis; neural network methods for genomics and transcriptomics; computational approaches to RNA structure prediction and differential expression; fairness-aware and reproducible ML for biomedical data; uncertainty quantification in biological deep learning models.

## Education

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### Rochester Institute of Technology

*Master of Science, Data Science — GPA 3.9*

Expected Aug 2026

*Rochester, NY*

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Bioinformatics Algorithms (BIOL-630), Neural Networks & Deep Learning, Data-Driven Knowledge Discovery (ISTE-780), Applied Data Science (DSCI-601), Foundations of Data Science (DSCI-633), Software Engineering for Data Science (DSCI-644).

### University of Rochester, Warner School of Education

*Graduate study, Teaching Computer Science K-12 (M.S. track) — GPA 4.0*

In progress

*Rochester, NY*

- NSF Robert Noyce Scholar (\$30,000 full funding); Advanced Certificate in Leadership in Disability and Inclusive Practices.

### Rochester Institute of Technology

*Master of Science, Computer Science — GPA 3.2*

2015

*Rochester, NY*

- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics.
- MS Thesis: ASL Recognition Application — image processing, feature extraction, and neural pattern classification.

### Farmingdale State College

*Bachelor of Science, Computer Engineering — GPA 3.3*

2012

*Farmingdale, NY*

- Dual degree: Electrical Engineering & Computer Engineering Technology.
- Honors: CSTEP Award & Merit; Dual BS Scholarship.

## Research and Professional Experience

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**Rochester Institute of Technology, Software Engineering Department**    Fall 2025 – Present  
*Graduate Assistant, Quantum and AI Research*    *Rochester, NY*

- Build reproducible ML and deep learning evaluation pipelines across local, Colab, and GCP environments, with structured logging and rigorous benchmarking methodology.
- Develop and evaluate neural network architectures for sequential and structured data, with emphasis on reliability, reproducibility, and generalization under distribution shift.
- Produce manuscript-ready technical analyses of deep learning methods including evaluation under noisy, heterogeneous, and partially observed data conditions.

**University of Rochester, Warner School of Education**    Sep 2024 – Present  
*CS Teacher Candidate & Education Research (NSF Noyce Scholar)*    *Rochester, NY*

- Deliver K–12 computer science instruction in active public school placement as part of the NSF Robert Noyce Scholar program, including computational thinking, inclusive pedagogy, and NYS learning standards.
- Conduct applied education research on AI-informed and data-driven approaches to improve learning outcomes for neurodivergent students, applying Universal Design for Learning (UDL) and evidence-based adaptive strategies.
- Produce lesson artifacts and instructional analyses documenting the intersection of AI, inclusive education, and evidence-based CS pedagogy.

**Rochester Institute of Technology**    2025 – 2026  
*RNA Structure Prediction and Biological Deep Learning (BIO614 and BIO550)*    *Rochester, NY*

- Led substantial computational work in BIO614: advanced RNA secondary structure prediction by integrating the Nussinov dynamic-programming algorithm with thermodynamic MFE enhancements using Turner parameters and ViennaRNA, achieving >90% accuracy on synthetic motifs.
- Extended prediction pipeline with LSTM and neural-network-based scoring components to capture long-range dependencies in RNA sequences beyond classical dynamic-programming scope.
- Produced formal manuscript-style project (*BIO614-FinalProjectProposal*, Overleaf) with reproducible implementation, structured evaluation, and biological interpretation.
- In BIO550: conducted RNA-seq differential-expression analysis, including reproducible preprocessing, quality-control validation (DESeq2 workflow), and biological interpretation of gene-expression patterns.

**Rochester Institute of Technology**    2024 – 2025  
*Equitable Bioinformatics Research — ISTE-780*    *Rochester, NY*

- Pioneered fairness assessment applied to RNA secondary structure prediction algorithms, implementing bias-detection frameworks using SHAP analysis and disparate-impact metrics.
- Developed 900+ line modular Python codebase for reproducible fairness-aware evaluation of bioinformatics ML methods.

- Demonstrated statistical significance ( $p < 0.01$ ) in algorithmic bias detection across demographic proxies in biological data.

**VEDADATA Inc.**

Sep 2022 – Aug 2024

*Data Solutions Engineer*

*New York, NY*

- Designed scalable Python data pipelines for ingestion, validation, statistical diagnostics, and analytics operations on large biological and health datasets.

**VIOME Inc.**

Jan 2021 – Sep 2022

*AI Data Solutions Engineer*

*New York, NY*

- Developed healthcare AI workflows including deep learning feature engineering, model experimentation, and evaluation for personalized health prediction in AWS environments.
- Built preprocessing pipelines for microbiome sequencing data, including normalization, QC filtering, and feature extraction for downstream ML models.

## Selected Technical Writing

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- *Big Data Medical Diagnosis: A Multi-Method Approach* (RIT M.S. Computer Science graduate research, 2015) — ensemble and deep learning methods for clinical classification on high-dimensional biomedical data.
- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report, 2025) — ensemble and regularized regression methods for clinical risk stratification under distribution shift.
- *BIO614-FinalProjectProposal* (Overleaf manuscript, 2026) — RNA secondary structure prediction with thermodynamic deep learning enhancements, reproducible benchmarking, and biological interpretation.
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025) — fairness-aware ML for genomic diagnostic algorithms with SHAP-based bias detection and statistical significance testing.
- *Differential Gene Expression in Murine DRG Neurons Following Sciatic Nerve Injury: An NGS Reanalysis* (BIOL550 Computational Biology Project, 2026) — reproducible bulk RNA-seq pipeline using DESeq2, QC validation, and pathway-level biological interpretation of nociceptor gene expression.
- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Proposal, 2025) — bandit learning with equity constraints applied to high-stakes clinical diagnostic decision-making, addressing demographic disparity in sequential clinical recommendation workflows.

## Awards & Recognition

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- NSF Robert Noyce Teacher Scholarship (2024–2026) — full funding for M.S. in Teaching Computer Science K–12.
- MS International Scholarship (2012–2015) — MESCYT & RIT, awarded for academic excellence.
- Honorable Mention Technology Award (2012) — NYS STEP/CSTEP, 2nd place capstone project.
- IEEE Alumni Member (2009–Present) — Member No. 90613000.

## Technical Competencies

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**Programming:** Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash

**Deep Learning & ML:** PyTorch, TensorFlow, Keras; LSTM, CNNs, Transformer architectures; scikit-learn, XGBoost; uncertainty quantification; SHAP analysis

**Bioinformatics:** ViennaRNA, Nussinov algorithm, Turner parameters, DESeq2, RNA-seq pipelines, BLAST, differential expression analysis

**Data and Statistics:** pandas, NumPy, SciPy, statistical modeling, reproducible benchmarking

**Infrastructure:** Git/GitHub, Docker, Linux, GCP, AWS, LaTeX

## Selected Fit for KTH Deep Learning for Biological Systems Position

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- Direct research experience applying deep learning and neural network methods to biological sequence and structure problems, including RNA structure prediction with thermodynamic enhancements (BIO614, Overleaf manuscript).
- RNA-seq differential-expression analysis background (BIO550) provides practical genomics pipeline experience relevant to downstream deep learning on transcriptomic data.
- Proven ability to produce formal, manuscript-quality research outputs in computational biology with reproducible implementations.
- Strong deep learning stack (PyTorch, TensorFlow, LSTM, CNN) with experience applying these architectures to biological and biomedical data beyond toy benchmarks.
- Industry experience (VIOME) building preprocessing and feature-extraction pipelines for microbiome sequencing data provides hands-on grounding in the challenges of real biological data at scale.